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QoS optimization through PBMR algorithm in multipath wireless multimedia sensor networks

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Abstract
Recent research in communication moves towards cyber space with modern applications such as multimedia surveillance, smart autonomous traffic control, advanced health care monitoring, industrial automation, and continuous remote environmental and structural monitoring. Such applications are extensively generates multimedia information and converse through Wireless Multimedia Sensor Networks (WMSNs). Path selection plays vital role in transmitting the multimedia data over wireless networks. Single path selection incurs routing failure due to the energy hole because of power shortage and traffic congestion. Due to energy hole in any particular node the packet gets dropped in the route from source to destination. The multipath routing algorithm selects more than one path to increase the rate of the packet delivery than the single path routing. The prediction of energy deficiency in sensor nodes is becoming over head in WSMNs. In order to improve the packet delivery and performance of the network, in this paper a Priority Based Multipath Routing Algorithm (PBMRA) is proposed. It consists of three modules: The Neighbourhood Selection identifies the neighbours; Multipath Construction constructs multiple paths from the indentified neighbours; and Priority Selection selects the high energized routes based on the priority of node attributes such as liveliness, path delay and hop count. The proposed algorithm is experimentally compared with other existing algorithms such as AODV, DSR, DSDV and OLSR. The experimental results show that the PBMRA algorithm increases the overall throughput, packet received rate and provides the reliable route than existing algorithms.

Keywords Network lifetime · Energy holes · Reliable routes · Path selection · Priority based multipath · WMSN

1 Introduction

Recent advancements in communication technology elevate the real world to cyber world through modern applications such as multimedia surveillance, smart autonomous traffic control, advanced health care monitoring and industrial automation. The task of continuous and distributed ambiance interaction and transfer of collected information to central infrastructure is performed by numerous tiny sensing devices in WMSNs. These multi-functional, low-cost, low-power miniature nodes perform this humongous task using Micro-Electro-Mechanical-Systems (MEMSs) and Complementary Metal-Oxide Semiconductor (CMOS) technologies [1]. The components of the WMSNs are shown in Fig. 1. WMSNs consist of sensor nodes which collect the information from the sensing region. The collected data is processed by the processing module. The processed data has been encapsulated and forwarded to the sink node using wireless module. The energy consumption is very high as a consequence of high bandwidth utilization for multimedia content [2]. From the communication perspective, timely delivery of the critical data

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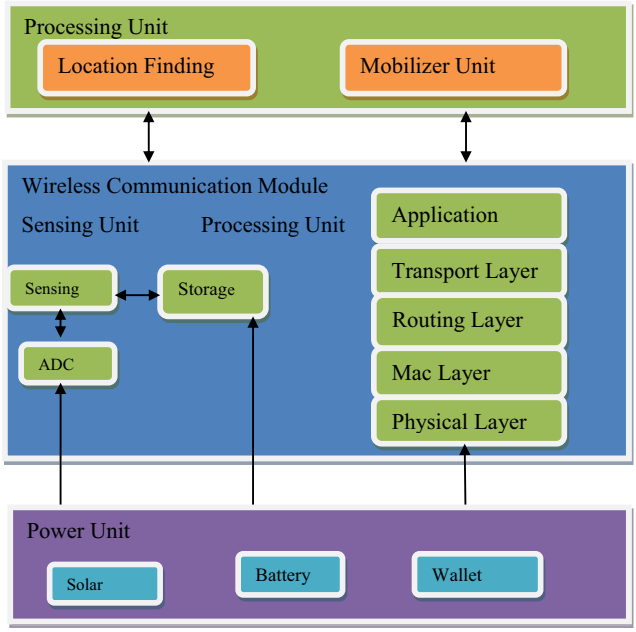


Fig. 1 Architecture of WMSNs

and strict adherence to Quality of Services (QoS) are the primary requirements of WMSNs. The routing path can be classified into single path and multipath. Single path routing learns the possible path and picked out a best path to reach destination. Though single path routing has smaller amount of process overhead, it is incapable for traffic load balance. Multipath routing learns the possible paths and can select optimal number of paths to the destination. Several research contributions exist for building multiple paths, and optimizing multipath data routing for WMSNs. The existing solutions solve the basic requirements of multipath routing over WMSNs. The major issues considered in multipath routing are mobile nodes, resource constraints of multimedia data and error prone shared broadcast radio channel.

Holes occur in WMSNs caused by deplete their energy level to process the data. Holes can be divided into two types such as static holes and dynamic holes. An energy fatigue dead sensor node creates static holes. Dynamic holes are formed when sensor nodes close together in a small area for the multimedia transmission. If dynamic holes occur in the routing path, it should be bypassed and additional routing paths are needed for increasing transmission performance. The multimedia streaming data huge in size and transmission packet over multimedia is always higher than the bandwidth of the sensor node.

Several approaches have been proposed for WMSNs routing in recent literatures. Existing approaches used single metric, either link quality or delay or energy level to create the multipath in WMSNs. This single metric produces infeasible multipaths in WMSNs. Some of the researches considered two metrics for while developing multipath routing algorithms. This paper presents a mutipath routing algorithms which considers three metrics such as hop count, delay, energy level to improve the packet delivery ratio. In this proposed

Table 1 Terminology used for PBMR algorithm

Parameter	Description
N	Number of nodes
G(V,E)	Directed Graph that represents sensor network topology
Sid	Source Id
Did	Destination Id
Lq	Link Quality
d_y	Delay
PRR	Packet Received Ratio
SNR	Signal Noise Ratio
RSSI	Received Signal Strength Indicator
ς	SNR
E_{dist}	Distance between two nodes
E i,j	Connectivity between nodes i and j
RDT	Route Discovery Time
NeTT	Net Traversal Time
NodeTT	Node Traversal Time
ND	Net Diameter
ϑ	Delay between source to sink
h_c	Hop count
Lq	Link Quality
Dy	Delay
χ	PRR
Ψ	Total Energy Level
Λ	RSSI
ψ	Packet Error Rate
BER	Bit Error Rate
BDF	Bits of Data Frame
B	Bandwidth in Hertz
NF	Noise Figure
RSS	Received Signal Strength
MEMSs	Micro-Electro-Mechanical-Systems
QoS	Quality of Service
NSA	Neighbourhood Selection algorithm
MCA	Multipath Construction Algorithm
PSA	Priority Selection Algorithm (PSA)
CMOS	Complementary Metal-Oxide Semiconductor
WSN	Wireless Sensor Networks
EEQAR	Energy Efficiency QoS Assurance Routing
ARCH	Adaptive Reliable Routing based on Cluster Hierarchy
ECMP	The Energy Constrained Multipath Routing
MCMP	Multi Constrained QoS Multipath Routing
DSR	Dynamic Source Routing
AODV	Ad Hoc On-Demand Distance Vector
DSDV	Destination Sequenced Distance Vector
OLSR	Optimized Link State Routing Protocol

Table 2 Neighbourhood selection algorithm

Neighbourhood Selection Algorithm
Begin
set n , Trp //No. of Nodes and Transmission Power
repeat
for each hop in the network
A disjoint node sends ***hi*** message to neighbourhood nodes that are in the Trp
for each neighbourhood node
if (el > threshold)
then reply RRep message to the disjoint node
end for
end for
until all nodes are visited in the network
end

method, we first use Neighbourhood Selection Algorithm (NBA) to establish the neighbourhood. Then a multipath construction algorithm (MCA) is proposed to construct the multipath from source to sink node. Priority selection algorithm (PSA) is proposed to provide the high energetic multipath based on the priority.

The remainder of the paper is organized as follows. The relative works of multipath routing is described in section 2 and section 3 deliberates the problem formulation. Section 4 discusses the proposed method. The section 5 discuss about the performance of proposed and existing algorithms. Section 6 concludes the paper with future direction.

2 Related work

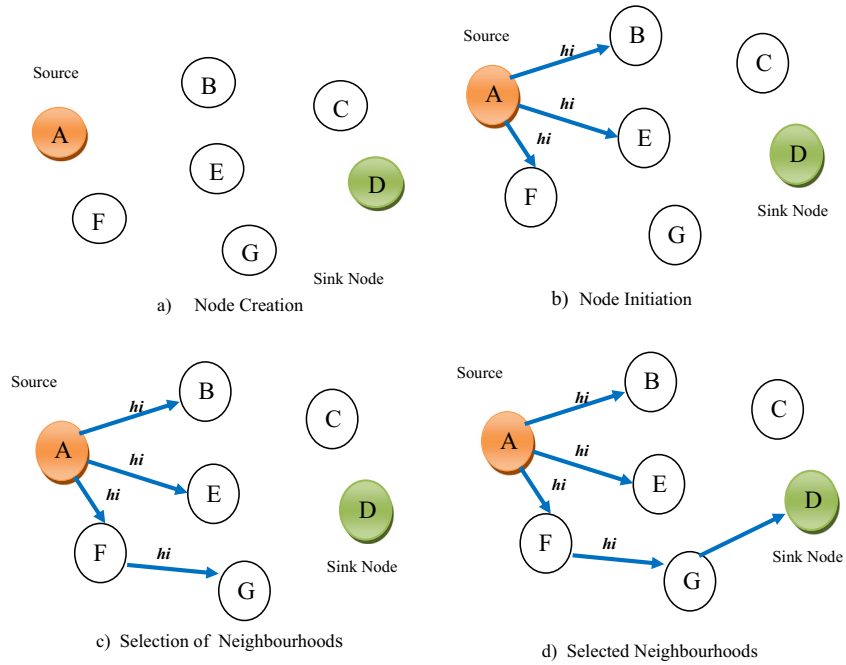
Mohammed Zaki Hasan et al. [3] prioritized the sensor nodes using only the link quality parameter which is calculated by Lagrangian relaxation method. The authors did not address about energy level of sensor nodes and delays. Vijay Ukani et al. [4] introduced adaptive power control method for efficient routing with energy constraint only. Incebacak et al. [5] proposed the Mixed Integer Programming to reduce energy consumption in Wireless Sensor Network and evaluated only two optimal paths from the multipath networks. Kai Lin et al. [6] identified the QoS trust estimation model to use cluster structure by cellular topology. EEQAR (Energy Efficiency QoS Assurance Routing) was compared with ARCH (Adaptive Reliable Routing based on Cluster Hierarchy) to improve the performance based on energy constraint alone.

Mariam AlNuaimi et al. [7] summarized the specifications of the different physical Layer Standards such as IEEE 802.15.4 ZigBee, IEEE 802.15.1 Bluetooth, IEEE 802.11 WiFi and 802.15.3a UWB. Luis Cobo et al. [8] proposed Ant Colony Optimization algorithm to improve the convergence and QoS. The performance of AntSensNet compared with the standard AODV in terms of delivery ratio, end-to-end delay and routing overhead. Lei Shu et al. [9] proposed TPGF (Two-Phase geographic Greedy Forwarding) geographic routing with planarization algorithm to provide the multipath routing but fails to predict the hole information in advance. Xiaoxia Huang et al. [10] proposed probabilistic modeling in multipath routing for wireless sensor networks. But their algorithm achieved higher redundancy by allowing similar packets over multiple paths and used local information. Yao Lan et al. [11] presented Shortest Path Algorithm to conserve energy and reduce the packet loss rate but fails in reliable transmission due to consider only two parameter energy and delay. Deb et al. [12] proposed randomized forwarding

Table 3 Control message *hi*

Sid	Did	h_c	d_y	l_q

Fig. 2 Neighbourhood selection



mechanism to decrease the error rate and overhead. But it unnecessarily considers all possible paths for data forwarding.

Bagula et al. [13] proposed Energy Constrained Multipath Routing (ECMP) algorithm to achieve efficient bandwidth utilization along with minimal usage of energy. But ECMP considered only energy level and not delay and link quality. Guangjie Han et.al [14] presented a detailed survey to compare with existing routing protocols for WMSNs for considering energy efficiency, reliability, data delivery model, hole bypassing, path latency, location awareness. A resource allocation process considered to the sensor node for improving the resource utilization in account of hop information. In order to maximize the network lifetime, tour and path planning method proposed by K., Almi'ani et al. [15]. The algorithm checks the remaining energy of individual nodes, by revisiting the entire network which leads to processing overhead. Y. Yang et.al [16] proposed the NC-RMR protocol to use hop-by-hop multipath to reduce routing broadcast and routing consumption by maintaining local routing information. But not focuses on priority of the multipath. M. H. Eiza et al. [17] proposed Ant colony optimization for energy efficient paths based on link quality only. A. Alanazi et al [18] proposed QoS routing protocols and analyzed different parameters such as energy aware, mobility, scalability, data aggregation and location awareness. But their proposed protocols didn't support scalability.

3 Problem formulation

The multipath routing with energy efficient sensors is represented with a directed network topology G = (V, E) Where |V| represents number of sensor nodes and |E| represents the number of edges in the network. The following Eq. (1) defines the distance between the two sensor nodes i and j.

$$E_{dist} = \sqrt{\left(x_i - x_j\right)^2 + \left(y_i - y_j\right)^2} \forall i, j = 1, 2, 3 \ldots, n \quad (1)$$

where (x_i, y_i) and (x_j, y_j) are the positions of ith and j^{th} node n. If there exists a direct link between two nodes then

$$E_{i,j} = 1 \quad (2)$$

Assume that each Edge is exemplified by three integer values: the energy level e_l; delay d_y; hopcount h_c. Every node on the network can update their e_l d_y h_c table periodically. The route exploration will be based on e_l, d_y, h_c value between source node to sink node. Net Diameter (ND) is used to compute the number of hops between two nodes in the network. Table 1 shows the terminolgy used for PBMR algorithm.

$$NeTT = 2 * NodeTT * ND \quad (3)$$

The average net traversal time is calculated using Eq. (3). The Route Discovery Time (RDT) is calculated using Eq. (4)

$$RDT = 2 * NodeTT \quad (4)$$

RDT is the estimation of maximum time needed to find the route in the network. The activities of sensor nodes are classified into two modes such as the wake up mode and sleep mode (Eq. (5)).

Table 4 Multipath construction algorithm

Multipath Construction Algorithm

```
//Source starts to establish multipath from source to sink.
Begin
    Set n, dy, eth
     update el dy hc table // energy delay hop count table
    set  Maxhopcount=5
    for every node i in the network
          visited.i = False
    end for
    for (i=1; i< Maxhopcount; i++)
        for(j=1; j<nodes_in_hop; j++)
            if(el.j >= threshold_energy) && dy.j < threshold_delay &&
            visited.j = False)
                Establish neighborhood
                Visited.j =True
                 break;
             end if
        end for
    end for
    construct topological view through el dy hc table
end
```

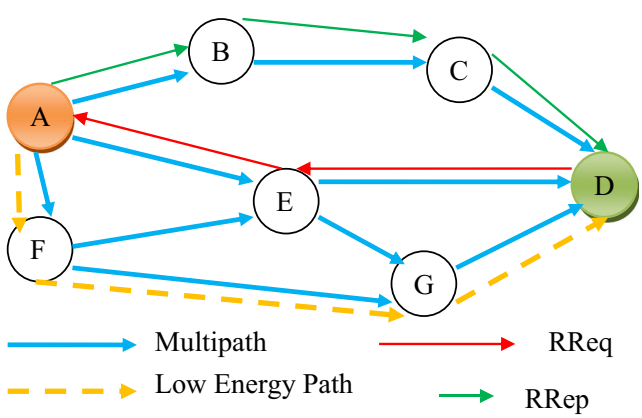


Fig. 3 Multipath construction

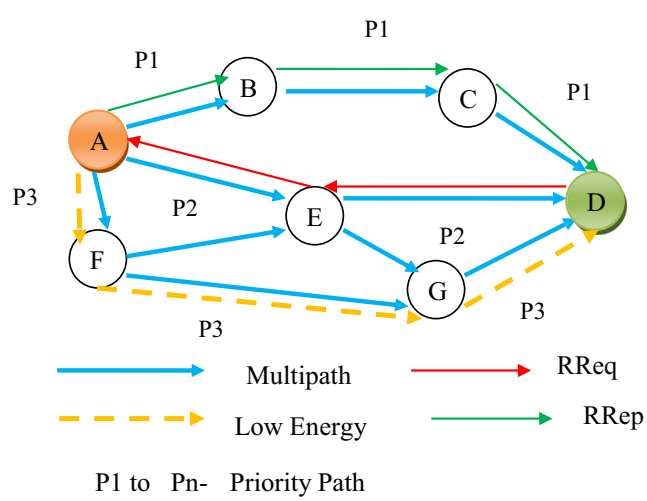


Fig. 4 Priority selection

Table 5 Priority selection algorithm

Priority Selection Algorithm
Begin
set Maxhopcount = 5
set sum = 0
for every path *p* in the network
for every node *i* in *p*
update e_l d_y h_c table
if(e_l >= threshold_enery)
set e_l.i_priority =1
if (d_y >= threshold_delay)
set d_y.i_priority = 0.5
if(h_c <=Maxhopcount)
set h_c.i_priority = 0.25
sum = sum + e_l.i_priority + d_y.i_priority + h_c.i_priority
end for
avg.p= Sum/3
end for
Sort the paths based on average value
End

The activity of sensor nodes

$$= \begin{cases} 1 \text{ if sensor node enters into wake up mode} \\ 0 \text{ otherwise sleep mode} \end{cases} \quad (5)$$

4 Priority based multipath routing algorithm (PBMRA)

A Priority Based Multipath Routing Algorithm is proposed in this paper to provide high energized multipath routing for packet transmission. The proposed algorithm consists of three modules: Neighbourhood Selection (NSA), Multipath Construction (MCA) and Priority Selection (PSA).The modules are explained in the following subsections.

4.1 Neighbourhood selection algorithm (NSA)

Initially a set of 'n' number of nodes is assumed to be deployed automatically in the network in Ad Hoc manner. After initial set up, every sensor node in the network invokes Neighbourhood Selection Algorithm to establish neighbourhood.

Table 6 Status of the sensor node

Time in Sec	Energy Level in J
3.61523	remaining energy = 4.30668
3.61549	remaining energy = 4.30644
3.6155	remaining energy = 4.30643
3.6157	remaining energy = 4.3062
3.622	remaining energy = 4.30104
3.62297	remaining energy = 4.29994
3.62298	remaining energy = 4.29993
3.62318	remaining energy = 4.29974
3.62323	remaining energy = 4.2997
3.62349	remaining energy = 4.29946
3.6235	remaining energy = 4.29945

Table 7 Energy level of the sensor Node

Time in Sec	STATE	Start Time in sec	Duration in sec
0.0676343	state = IDLE	start = +67,624,264.0	duration = +10,066.0
0.0679383	state = RX	start = +67,634,330.0	duration = +304,000.0
0.0684883	state = IDLE	start = +67,938,330.0	duration = +550,000.0
0.0684883	state = TX	start = +68,488,330.0	duration = +704,000.0
0.0692024	state = IDLE	start = +69,192,330.0	duration = +10,066.0
0.0695064	state = RX	start = +69,202,396.0	duration = +304,000.0
0.0695564	state = IDLE	start = +69,506,396.0	duration = +50,000.0
0.0704844	state = RX	start = +69,556,396.0	duration = +928,000.0
0.0704944	state = IDLE	start = +70,484,396.0	duration = +10,000.0
0.0704944	state = TX	start = +70,494,396.0	duration = +304,000.0
0.0709084	state = IDLE	start = +70,798,396.0	duration = +110,000.0
0.0709084	state = TX	start = +70,908,396.0	duration = +8,704,000.0
0.0796225	state = IDLE	start = +79,612,396.0	duration = +10,066.0
0.0799265	state = RX	start = +79,622,462.0	duration = +304,000.0

The Neighbourhood Selection algorithm is given in Table 2. It is used to find neighbours, make relationship with neighbour and establish connection between them. Then, the neighbourhoods table is updated with their unique Neighbour ID.

The algorithm is explained with an example (Table 3). Initially the nodes A, B, C, D, E, F and G are deployed in the network (Fig. 2a). The node A initiates the neighbourhood selection process by constructing the control message ***hi*** (Table 3) which consist of Sid, Did, hc, dy, lq. The nodes who receive the control message ***hi*** will check their hc, dy, lq value with received message. The one who fulfils the conditions of ***hi*** message will establish connection with source node A and update its table.

Table 3 shows the structure of the control message ***hi***. It consists of source id (Sid), destination id (Did), hopcount h_c, enegy level e_l, delay d_y, and link quality l_q where Sid represents the source, Did represents the destination, hc hop count says how far the nodes are located and how could make possibilities of data transmission between them and delay field represents the time taken to transfer the data from node n_i to n_{i+1}. The sum of the link delays is given in Eq. (6) .

$$\vartheta = \sum_{i=1}^{S} d_y(n_i, n_{i+1}) \tag{6}$$

Where $d_y(n_i, n_{i+1})$ is the delay of data over the node ni and n_{i+1} and S denotes the sink or destination node. Energy Level denotes the remaining energy of the nodes lie in the path. The total energy level of the network [13] is given by

$$\Psi = \sum_{i=1}^{S} e_l(n_i, n_{i+1}) \tag{7}$$

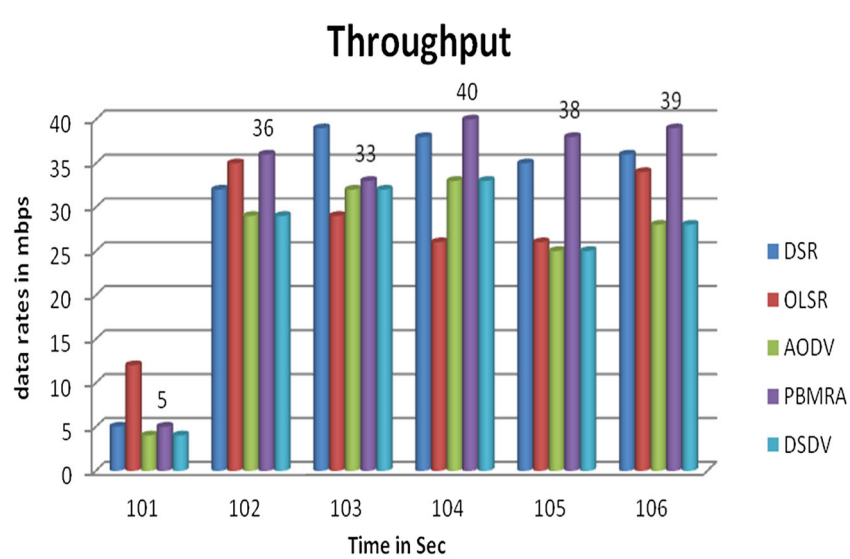


Fig. 5 Throughput

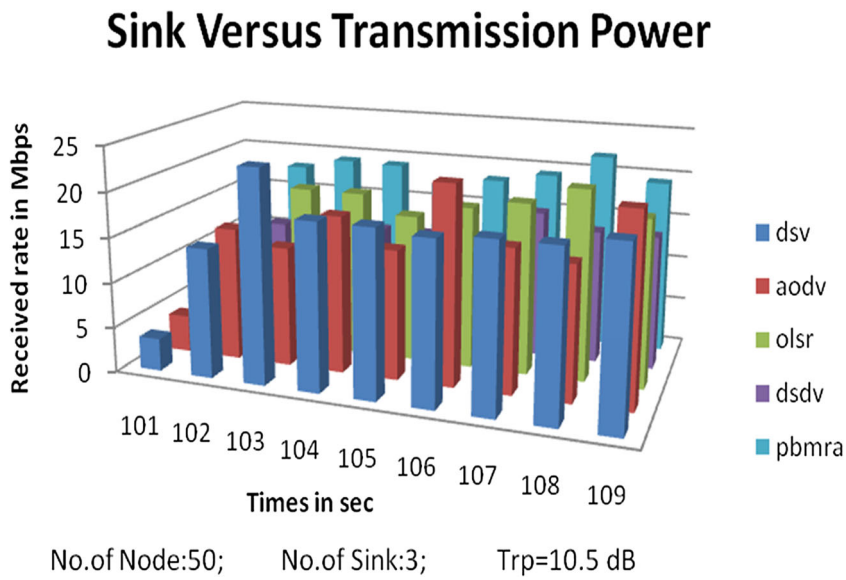


Fig. 6 Sink versus transmission power

where e_l (n_i, n_{i+1}) is the energy level between the nodes n_i and n_{i+1}.

The Signal to Noise ratio is calculated using Eq. (8)

$$\zeta = Trp/Trn \tag{8}$$

where Trp denotes transmission power and Trn denotes received power. Link Quality differs over space and time [19–21]. Equation (8) is applied to forecast the link quality (lq) [22, 23] between the node and its neighbours. Every sensor node knows the link quality of its neighbouring nodes [24, 25].

The packet error rate is calculated using Eq. (9)

$$\psi = \left(1-(1-BER)^{bdf}\right) \tag{9}$$

where BER = Total Number of error bits / Total Number of bits [26].

The Packet Receive Ratio (χ) is the ratio of the total received packets and the transmitted packets.

$$\chi = 1/\lambda \sum_{i=1}^{n} P_R \tag{10}$$

where λ is the number of transmitted packet and P_R is the number of received packets. RSSI(Λ)is the sum of the pure received signal and the noise floor at the receiver and is calculated using eq. (11).

$$\Lambda = \sum_{i=1}^{n} Rss + Nf \tag{11}$$

The noise floor can be measured by the summation of all the noise sources and undesirable signals where undesirable signals are made within the complete data acquisition and signal processing system.

$$N_f = -174 + NF + 10\log B \tag{12}$$

where NF (Noise Figure) = 10 * log (F), Noise Factor F = (Tn /290) +1 and B is bandwidth.

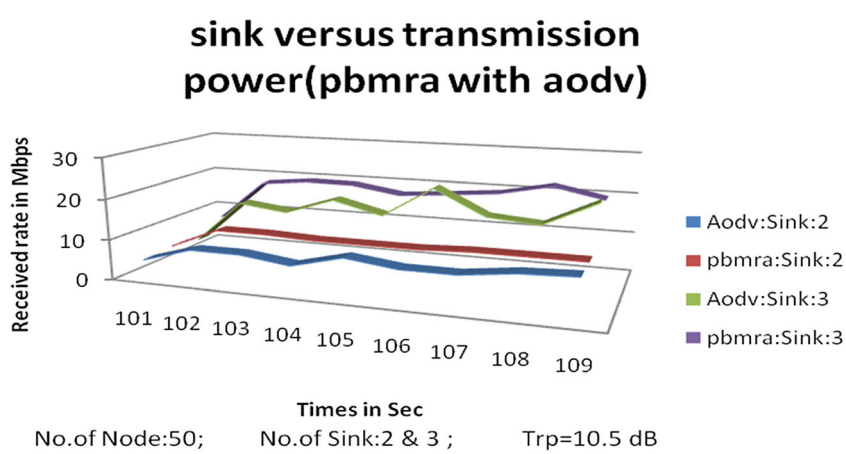


Fig. 7 Node versus sink node

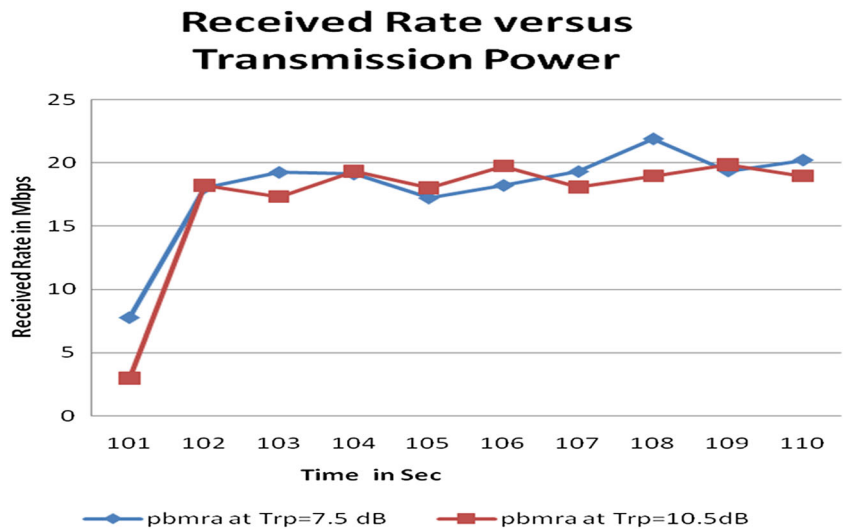


Fig. 8 Received rate versus transmission power

4.2 Multipath construction algorithm

Multipath routing is needed to improve network performance under heavy traffic conditions. After neighbour relationships are established, electing possible paths and choosing best paths from the source to sink is mandatory to transmit multimedia data. Usually the routing algorithm runs on every node on the network. Discovering and establishing paths are mandatory that is initiated by sink to the source. The multipath construction algorithm is given in Table 4.

For every sensor node threshold value is determined depending on the energy level. Sink node sends Rreq control packet to the source node via intermediate active nodes. The active nodes started to update their e_l, dy, h_c table. Also they cross check their e_l, dy, h_c table status with immediate neighbors. Table 4 shows how the multipath is constructed from sink to the source. Node D (Sink node) would like to collect the information from source node A, so it floods the control packet Rreq via Intermediate node B, C, F, E, G (Fig. 3). After receiving Rreq from the sink node D, Node A will send the reply control packet, RRep, to the sink D. After handshaking with each other the multipath will be explored from the node A to the node D shows in Fig. 3.

4.3 Priority selection algorithm

In WMSN, multiple paths [27] is available to route data from the sink to source. Discovering multiple paths in the network is a critical process. Even if multiple paths are discovered, it may have energy dearth nodes or it may have excess delay which leads to the problem. Hence some prediction based decision to be made to overcome this scenario.

The identified multiple paths are prioritized using e_l, dy, h_c values shown in Fig. 4.

The priority of each path is calculated using the average value which is given in Table 5. The path that has the highest average value will received first priority and that will be treated as high energized routing path and the next average value of the path will be the next energized path and so on from

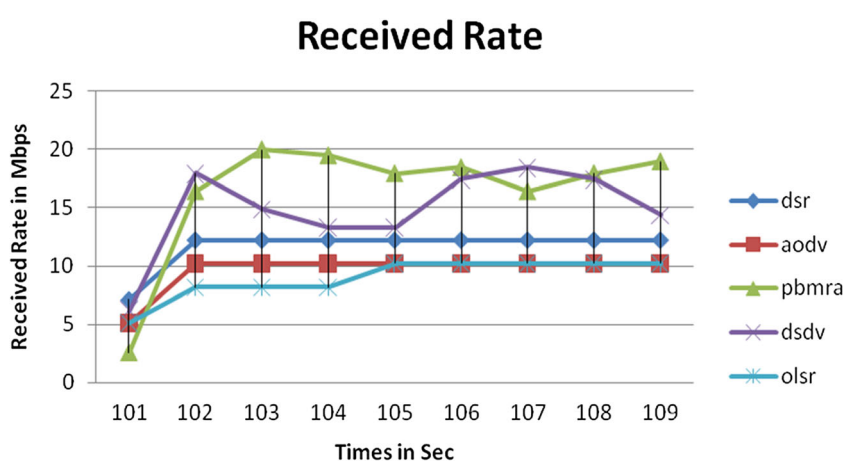


Fig. 9 Received rate

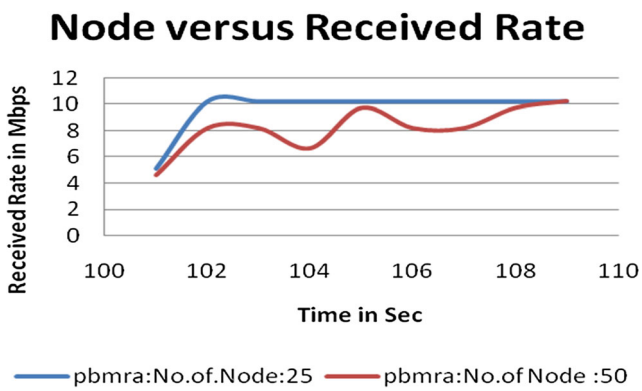


Fig. 10 Received rate

source to sink [28]. The low energy level path will be discarded automatically [29].

4.4 Hole information

Hole is represented as static hole and dynamic hole. Energy depletion plays major role in reduction of network performance. When considering the network load, AODV, DSV, DSDV, and OLSR generate almost the same network load at the beginning of the simulation. Afterwards, the routing paths are broken due to depletion of their energy level. Hence, hole should be identified before constructing the multipath from source to the sink. Based on the energy level of the sensor node, the multipath is constructed by PBMRA. Tables 6 and 7 show the status of the sensor nodes and energy level [29] of each sensor node in the network. PBMRA identifies both the static hole and dynamic hole. The multipath construction algorithm has chosen energized multipath from source to the sink because holes are identified well in advance.

5 Results and discussions

All the algorithms are implemented in NS3 simulator. The simulation is assumed on a uniform topology deployed of 50 nodes spread in a square area of 500 m * 500 m in the Wireless LAN Wi-Fi Physical Standard IEEE802.11.b. The transmission range is 2.4GHz, 54 Mbps. Initially the transmission power of all nodes is 7.5 dBm. The delay is evenly distributed between 0 to 200 s and interval of 1 s.

Table 8 Comparison of algorithms' characteristics

Characteristics	DSDV	DSR	AODV	OLSR	PBMRA
Multicast	No	No	Yes	No	Yes
Periodic broadcast	Yes	No	Yes	Yes	Yes
Reactive	No	Yes	Yes	No	Yes

The delay represents the time required for queuing, transmission, retransmission, contention and propagation of packets.

The proposed PBMR algorithm is compared with existing DSR, AODV, DSDV, and OLSR algorithms interms of throughput, received rate and delay. The simulation results give the better throughput compared with other algorithms and their results are plotted shown in Fig. 5. The PBMRA gives 0.77% increased throughput than AODV, 0.99% than DSR, 0.87% than OLSR and 0.79% than DSDV. The obtained results say the PBMRA is stable and provides the better result compare to other foresaid protocols for multipath wireless multimedia sensor network environment.

The received rate of PBMRA is compared with the existing algorithms with respect to different parameters such as sink versus transmission power, node versus transmission power, received rate versus transmission power, and node versus received rate. The first scenario considers a set of 50 nodes, 2 sink nodes and 10.5 dbm transmission powers to forward the packets in the multipath. The second scenario considers a set of 50 nodes, 3 sink nodes and 10.5 dbm transmission powers to forward the packets in the multipath.

Figure 6 depicts the sink versus the transmission power for PBMRA algorithm with DSR, AODV, DSDV, and OLSR. The energetic sink node is needed to increase the performance of the networks. The results show the sink versus transmission power for 50 nodes, 3 sinks and transmission power of 10.5 dbm. As the number of sink nodes increases received rates also increases based on the transmission power.

Figure 7 portrays sink versus transmission power for PBMRA with AODV. The comparison results show the sink versus transmission power for PBMRA with AODV. There is no significant changes of received rate in PBMRA with AODV compare with first and second scenario. The received rate of two scenario is approximately 0.86%. Figure 8 shows the received rate of PBMRA at Trp = 7 .5 db and PBMRA at Trp = 7.5 db and the transmission power increases as the received rate is also moderately increases.

The received rate of PBMRA is significantly improved than other routing protocols as shown in Fig. 9. Similarly the analysis of the network is extended to node vs. received rate as shown in Fig. 10. For analyses, 25 and 50 nodes are taken into account. The result shows that for the minimum number of nodes, the PBMRA maintains the consistent network throughput is consistent and obtains maximum data rate. The increase in nodes in the network share bandwidth which results in decrease in data rate consequently. Table 8 represents the comparison of the characteristics of an algorithm such as multicast, periodic broadcast and reactive.

For a Given network G(V,E), the time Complexity of proposed Multipath Routing algorithm is O(2d) * O(p log l) where d is Network Diameter and p is number of routing paths, l is number of links.

6 Conclusion

Wireless Multimedia Sensor Networks (WMSN) severely suffers from energy dearth nodes and excess delay in packet delivery. In this paper, a Priority Based Multipath Routing (PBMR) algorithm is proposed for efficient routing of packets in WMSNs. The proposed algorithm constructs multiple paths, prioritizes them and finally identifies the high energized path. The algorithm also identifies the energy hole in advance. The proposed PBMRA is experimentally compared with existing AODV, DSV, DSDV and OLSR protocols with respect to the following metrics: packet received ratio, delivery delay and throughput. The PBMRA gives 0.77% improvement in throughput against AODV, 0.99% against DSR, 0.87% than OLSR and 0.79% than DSDV. From the results it is observed that the PBMRA is stable and provides better packet delivery ratio and minimum delivery delay compared to foresaid protocols for multipath wireless multimedia sensor networks. In PBMR algorithm, the delay increases as the number of nodes increases. In future research work, we will try to reduce the delay and will focus on optimal routing technique for WMSN.

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